CASE STUDY 05 / 05

Data Stewardship & Quality Controls

Master-data remediation • data dictionary • issue ownership and lineage

BUSINESS SCENARIO

A fictional B2B organization consolidates CRM customer records before a reporting launch. The steward must define canonical data, review duplicates, establish valid-value rules, route missing ownership, and publish accountable data-quality metrics.

COMPLETED DELIVERABLE Governed data dictionary, duplicate-match adjudications, canonical-golden-record sample, risk/exception log, KPI definition sheet, and stewardship SOP.

ROLE COVERAGE

Data Steward • Data Processor • Data Verification Specialist

EVIDENCE AT A GLANCE

480	462	100%
SOURCE ROWS	GOLDEN RECORDS	MATCHES REVIEWED

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Portfolio disclosure: All organizations, records, people, and transactions are fictional. This is a simulated work product, not a claim of past paid employment or client work.

01 / GOVERNANCE CHARTER

Dataset contract and control ownership

Element	Definition
Dataset	CRM_CUSTOMER_MASTER / release CRM-R2026-10-27
Business owner	Head of Revenue Operations (fictional)
Steward	Customer Data Steward — accountable for semantics and issue triage
Technical custodian	CRM Systems Administrator — responsible for access, transport, backup
Input population	480 source records across sales CRM and onboarding queue
Published population	462 canonical master records after 18 confirmed duplicate removals
Excluded content	No real customer records, private contact details, credentials, or live system exports

Stewardship principles

- Every master record has a stable master key and traceable source keys.
- Match scores produce review candidates; merging requires evidence and a recorded adjudication.
- Business definitions and permitted values are documented and versioned, not embedded only in an undocumented script.
- Data issues have an owner, severity, due date, and measurable closure evidence.

02 / DATA DICTIONARY

Approved canonical field dictionary

Field / type	Meaning	Required	Rule / owner
customer_id / TEXT	Permanent master identifier	Yes	CUST-#####; CRM custodian
source_keys / TEXT	Original source references	Yes	One or more immutable IDs; steward
legal_name / TEXT	Registered entity name	Yes	Do not expand abbreviations without evidence
display_name / TEXT	Customer-facing preferred name	No	May differ from legal name; sales owner
primary_email / TEXT	Authorized operational contact	Yes	Lowercase; valid pattern; steward
state_code / CHAR(2)	US region of main location	Yes	Allowed US state / DC codes
account_owner / TEXT	Responsible sales user ID	Yes	Must map to active owner directory
lifecycle_status / ENUM	Current customer state	Yes	Prospect / Active / Suspended / Closed
last_verified_at / DATE	Date verified by source of truth	Yes	YYYY-MM-DD; no future date
record_source / ENUM	Primary provenance identifier	Yes	CRM / Onboarding; custodian

DATA CONTRACT

A contact address is not a unique customer identity: equal emails can indicate a likely match, but legal-name and operational evidence still determine final merge disposition.

03 / MATCH ADJUDICATION

Potential duplicate review log (illustrative cases)

Pair	Candidate evidence	Decision	Control rationale
S-019 / S-204	Normalized legal name and shared domain	Merge	Same account ID confirmed in onboarding case
S-061 / S-062	Exact phone; two distinct legal entities	Keep both	Shared reception line is not identity proof
S-097 / S-371	Minor punctuation variant and exact source tax-key hash	Merge	Verified source registry match (synthetic hash)
S-155 / S-157	Similar name; distinct owner/customer contracts	Keep both	Insufficient evidence; false match
S-222 / S-228	Same entity, one legacy alias	Merge	Alias and legal entity mapped by approved owner

Full-batch disposition summary

Classification	Count	Interpretation
Candidate duplicate pairs reviewed	27	All reviewed by steward
Confirmed duplicate rows consolidated	18	18 removed from source count; lineage preserved
False-positive candidate pairs retained	9	Records remain separate
Canonical customer records released	462	480 input − 18 consolidated = 462

REVIEW CAUTION

The table shows representative adjudications, not all 27 source-pair investigations. All 27 simulated candidate pairs have a recorded disposition in the summary. These are constructed training results, not measured production-system performance.

04 / GOLDEN RECORDS AND EXCEPTION ROUTING

Approved canonical-master sample

Master ID	Contributing source	Legal name	State	Status
CUST-00019	S-019 + S-204	Marlow Business Supply	NC	Active
CUST-00061	S-061	Willow Ridge Partners	GA	Active
CUST-00062	S-062	Willow Ridge Holdings	GA	Prospect
CUST-00097	S-097 + S-371	Stonebrook Analytics	VA	Active
CUST-00155	S-155	Solace Health Equipment	SC	Active
CUST-00157	S-157	Solace Medical Services	SC	Active
CUST-00222	S-222 + S-228	Briar & Finch Creative	NC	Active

Open issues by owned queue

Ticket	Control failure	Owner	Count	Action
DQ-021	Missing active account owner	Sales Ops	10	Assign / verify owner mapping
DQ-022	Invalid or unknown state code	CRM Admin	3	Verify authoritative geography
DQ-023	Unconfirmed account alias (nonblocking)	Data Steward	1	Retain original until verified

The 13 blocking records (10 missing owner, 3 invalid state) are distinct in this simulated snapshot. The separate one-record alias question is a nonblocking review item and does not alter those 13 counts.

05 / QUALITY KPIS & LINEAGE

Quality scorecard — published snapshot

Metric	Calculation	Result	Target
Source-to-master reconciliation	480 − 18 = 462	462 rows	Exact
Owner completeness	452 / 462 nonblank valid owners	97.84%	≥ 99.0%
State validity	459 / 462 valid codes	99.35%	≥ 99.5%
Duplicate decisions	27 / 27 candidate pairs adjudicated	100.00%	100%
Unresolved blocking exceptions	10 owner + 3 region items	13	0

Data lineage narrative

01 Source acquisition: sales CRM and onboarding export (batch snapshots) are ingested with original source keys intact.

02 Validation: required-field checks, code-list checks, reference-owner matching, and candidate duplicate detection.

03 Steward adjudication: each candidate pair receives Merge or Keep with a reason and evidence key.

04 Canonicalization: resolved records receive a stable master ID and an array of contributing source IDs.

05 Publication: release version stamped and quality scorecard attached; unresolved errors remain in owner queues.

INTERPRETATION

The released dataset contains 462 master records but falls below the stated owner-completeness and state-validity targets. It is explicitly a conditional release for noncritical internal reporting; downstream operational uses requiring complete ownership are blocked until defects are resolved.

06 / OPERATING MODEL & RELEASE

Stewardship operating procedure

01 Maintain a versioned business glossary and approved field dictionary.

02 Ingest immutable source snapshots and capture the extraction time, origin, and row count.

03 Validate required fields and reference lists; report population-based denominators.

04 Generate duplicate candidates and review evidence with a two-person approval for irreversible merges.

05 Publish canonical mapping plus source-key lineage; never discard adjudication evidence.

06 Assign each open issue a severity, accountable owner, due date, and closure test.

07 Publish dashboard measures with explicit definitions, as-of date, and permitted use.

08 Re-run controls after corrections and require a change log and signoff before the next release.

Ownership and escalation matrix

Issue	Day 1	Day 3	Day 5
Missing account owner	Assign Sales Ops ticket	Manager follow-up	Escalate to RevOps lead
Invalid state code	CRM Admin checks source	Source owner confirms	Escalate as data contract breach
Disputed merge	Steward opens evidence review	Second-person adjudication	Refer to business data owner

CONTROLLED RELEASE

Control	Final result
Document owner	Enterprise Data Governance / Data Steward
Review state	Conditional internal reporting release; 13 blocking defects remain queued
Review date	2026-10-27
Data classification	Fictional sample / no real personal, financial, or medical data

07 / REPRODUCIBLE EVIDENCE

480-row source and 462-row canonical master

This version replaces an aggregate-only simulation with a complete reproducible synthetic data pack. All 480 source IDs, 462 published master entries, 27 pair-level decisions and 14 issue tickets are supplied as UTF-8 CSVs. No real accounts or customers are included.

Population control	Recomputed count
Source batch rows	480
Confirmed duplicate rows merged / preserved in lineage	18
Canonical master rows released for limited use	462
Reviewed candidate pairs	27 = 18 merge + 9 keep
Distinct blocking records	13 = 10 owner + 3 invalid region
Additional nonblocking alias review	1

Final quality metrics, based on companion CSV

Metric	Formula	Actual	Target
Owner completeness	452 / 462	97.84%	99.00%
Valid state	459 / 462	99.35%	99.50%
Candidate decision completeness	27 / 27	100%	100%
Source lineage coverage	480 / 480	100%	100%

08 / GOVERNANCE AUDIT

Lineage and exception traceability

Companion file	Operational demonstration
SOURCE_customer_snapshot_480.csv	Original 480 rows with source IDs and immutable external keys
FINAL_canonical_master_462.csv	462 unique IDs; contributing-source lineage retained
DECISIONS_all_27_candidate_pairs.csv	18 merge decisions plus 9 false-positive keep decisions
EXCEPTIONS_owned_queue_14.csv	10 owner issues, 3 state defects, 1 alias review

Independent quality-review sequence

- Count input source IDs and confirm 480 unique keys.
- Count canonical masters and confirm 462 unique keys; all 480 original IDs appear in contributing-source lineage.
- Review every decision in the 27-pair ledger and verify that false positives remain distinct.
- Measure owner completeness and state validity with denominators of 462; explicitly fail the stated 99%/99.5% thresholds.
- Keep conditional internal reporting separate from unrestricted operational use until 13 blocking defects are fixed.

Important: the original case study displayed representative adjudications only; this evidence edition includes the complete candidate decision ledger. All counts are synthetic, measured only for this constructed exercise.

DEEP-DIVE CASE FILE / STEW-01

Governance operating model and rule catalog

The original 480-row source / 462-row canonical analysis remains the primary completed scenario. The additional control catalog converts narrative quality expectations into testable rules with owners, thresholds, escalation and review evidence. Targets are internal training policies, not claimed company service levels.

ID	Dimension / field	Target	Breach routing
DQ-01	source_id / Uniqueness	100%	block if duplicate
DQ-02	legal_name / Completeness	>=99%	queue if missing
DQ-03	contact_email / Syntax & completeness	>=98%	hold invalid email
DQ-04	state / Reference validity	>=99%	hold unknown code
DQ-05	account_owner / Ownership completeness	>=95%	assign pending owners
DQ-06	external_key / Reference uniqueness	100% canonical	investigate collision
DQ-07	parent_source_id / Lineage validity	100% linked refs	block orphan links
DQ-08	source_status / Approved vocabulary	100%	correct controlled status

RACI / release ownership

Task	Data processor	Data steward	Control owner	System owner
Raw ingestion	R	C	I	A
Duplicate adjudication	C	A/R	C	I
Quality metric signoff	R	R	A	I
Canonical release	R	R	A	C
Access control review	I	C	A	R

R = Responsible; A = Accountable; C = Consulted; I = Informed. This is a fictional team operating model demonstrating governance separation of duties.

DEEP-DIVE CASE FILE / STEW-02

Lineage, retained evidence and data incident procedure

- Capture source_snapshot with immutable file hash, extraction time and schema version.
- Profile 480 source rows without altering them; record completeness and proposed duplicate pairs.
- Adjudicate 27 candidate pairs with approved match criteria, not just fuzzy-string similarity.
- Build the 462-row canonical export from approved decisions; keep removed/merged source links.
- Record 14 exceptions with owner, next action and follow-up cadence.
- Validate unique business keys, required attributes and reference integrity before publishing.
- Store QA test evidence, approval identities, and release change log with an explicit retention period.

Breach response - mock incident

Minute	Action	Evidence
0-15	Detect unusual spike in unowned records	Quality-rule alert / batch ID
15-30	Pause canonical release; preserve raw snapshot	Release hold and immutable source
30-60	Route to control owner; identify affected rows	Issue log / scope estimate
60-120	Correct controlled values; rerun checks	Change history / re-validation
After fix	Obtain reviewer signoff and document prevention	Release approval and root-cause note

DEEP-DIVE CASE FILE / STEW-03

Reviewable quality tests and SQL demonstration

Test	Method	Expected
ST-01	Count raw / final files	480 raw; 462 canonical
ST-02	Inspect 27 match decisions	All pairs show rationale and outcome
ST-03	Review 14 open exceptions	Each has accountability evidence
ST-04	Find duplicate canonical external keys	Investigate any collisions
ST-05	Find missing account owners	Queue and assign owner
ST-06	Check parent linkage against source IDs	No orphan released link
ST-07	Recalculate metric definitions	Numerator, denominator, scope repeatable
ST-08	Inspect export row lineage	Each canonical row traces to source

The companion SQL_data_quality_review_examples.sql contains dialect-neutral review query templates for canonical collisions, missing owner information, parent-key integrity and source-to-target totals. They are examples for demonstration; no query execution against a live database is claimed.

Stewardship judgment: do not over-merge

A shared contact address, similar organization name, or near-match external key is a candidate, not proof of identity. Confirm the governed matching threshold, preserve the source IDs, record an auditable rationale, and avoid deletions without approval.

Interview narrative

“I built a simulated quality-governance workflow around 480 source rows and 462 published records. I documented 27 candidate matching decisions, 14 assigned exceptions, a data dictionary, rule thresholds, escalation paths and repeatable lineage checks.”